CCCA Cell-State Analysis — Qian et al. 2020, Ovarian

Ovarian Cancer · 10x · 6,393 malignant cells (4 samples)

GO / NO-GO GATES

Gate 1 — mixing & continuity **GO**

Harmony(theta=20) interleaves all 4 patients: same-sample neighbour fraction 1.00 -> 0.63 (chance 0.35), iLISI median 1.00 -> 2.13 of 4. theta=30 REVERSES mixing to 0.88 (over-integration fragments into per-sample lobes), so theta=20 is the strongest integration on the plateau. 97 cells / 1 detached island removed -> one connected component. Read on the regenerated embedding + iLISI: mixed AND continuous.

Gate 2 — module geography **GO**

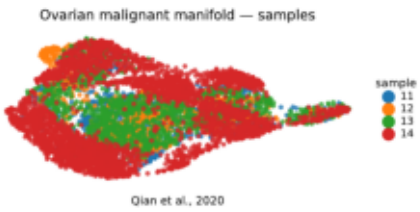
Non-branching case: a single (Gast/pSom)->Crest developmental gradient. Gast & pSom co-express (joint enrichment 1.20 > 1; smoothed r=+0.14), and both oppose Crest (Gast<->Crest r=-0.28, pSom<->Crest r=-0.26; joint enrichment 0.58 / 0.63 < 1 = inverse, not a co-expression hub). Along pseudotime Gast/pSom fall (rho -0.46 / -0.55), Crest rises to a terminal tip (high-decile pt = 0.83).

Gate 3 — cross-repression **NA**

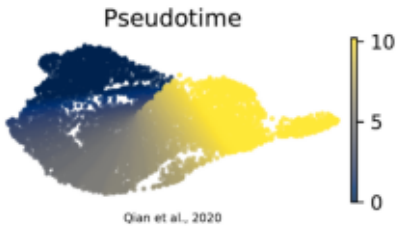
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks, fold self-approach 2.41). No bifurcation, so the branching-only hub cross-repression test is not run (the correlation / modules notebooks are not produced, per the deliverable spec).

STEP FIGURES

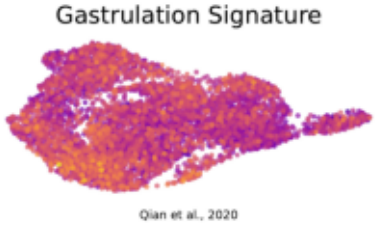
01 · Samples (Gate 1)



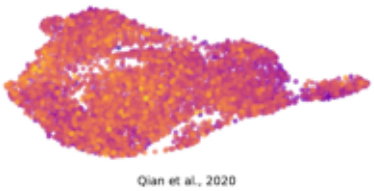
02 · Pseudotime



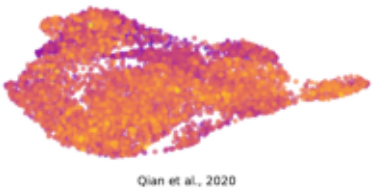
03 · Gastrulation



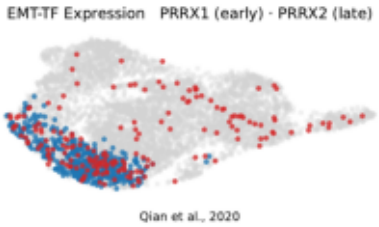
03 · Somitogenesis



03 · Neural Crest



04 · EMT-TFs



Dataset: Ovarian cancer (Qian et al. 2020), 10x, 6,393 malignant cells over 4 treatment-naive patients; one continuous low-confound body (cycling gradient, embryonic modules proliferation-independent).

Verdict: Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation/somitogenesis -> neural-crest axis, Crest terminal. G1 GO / G2 GO / G3 NA. EMT-TFs PRRX1 (early/root) -> PRRX2 (late/terminus).

PARAMETER CHOICES

Sample threshold	>=100 cells (floor; no dominant multiplicative cliff, top/2nd gap ratio 1.89) -> 4/4 samples, 6,490 cells
Overdispersed genes	1,013 (scFates find_overdispersed)
Integration	Harmony on sample, theta=20 (mixing plateau; theta=30 over-integrates and fragments)
Embedding	n_pcs=20, n_neighbors=30, UMAP spread=1.3 (10x precedent scale); 97 cells / 1 island removed -> 6,393
Geometry	Non-branching -> scFates principal curve (ElPiGraph), Nodes=20, epg_lambda=0.3, epg_mu=100
Topology check	2 tips / 0 forks; fold self-approach 2.41 (>1 fold-free); corr(x_lin, pseudotime)=+0.83
Root	tip 0 = Gast/pSom-high, Crest-low developmental origin (derived from signature order)
Module means (raw)	Gast +0.042 · pSom +0.065 · Crest -0.019 (Linear-pSom/Crest)
EMT-TFs shown	PRRX1 (early/root, rho -0.19) <-> PRRX2 (late/terminus, rho +0.00); detection-gated, FOXC/PRRX in panel